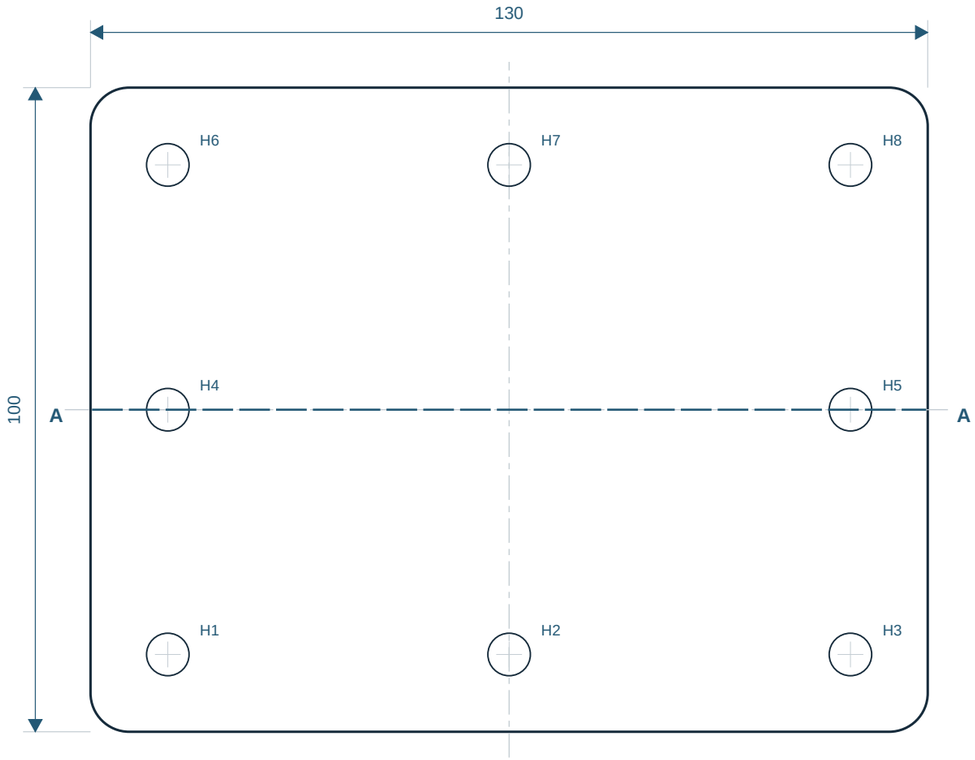
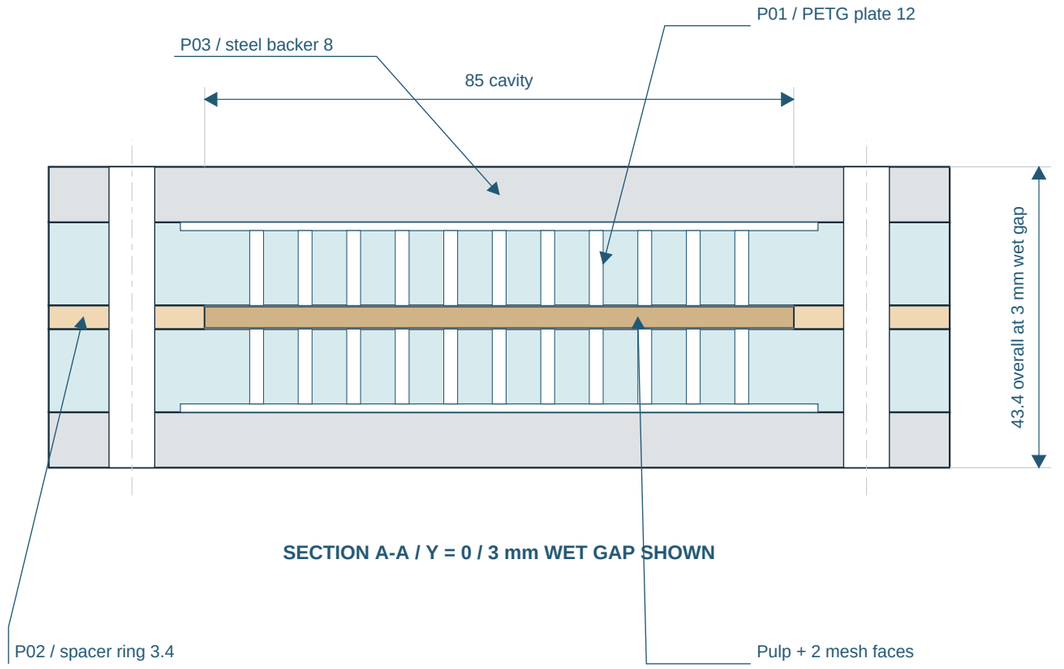
# ASSEMBLY / VARIABLE WET THICKNESS




TOP VIEW / STEEL BACKER



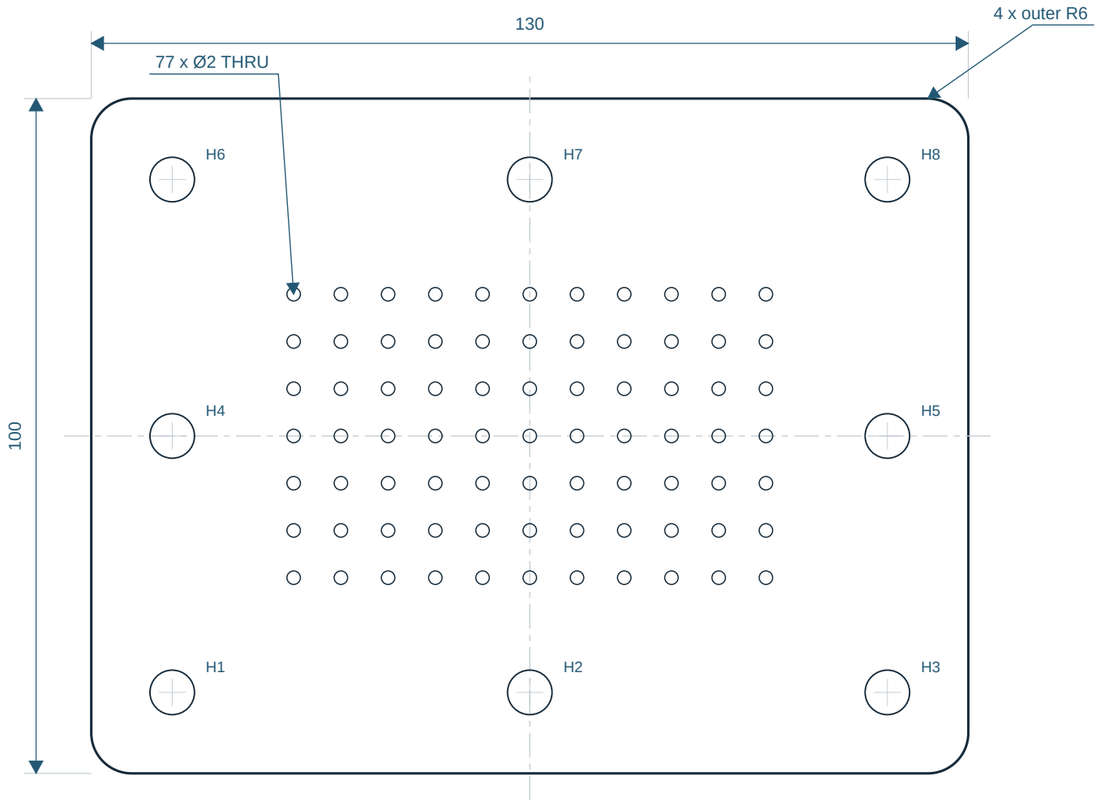

SECTION A-A / Y = 0 / 3 mm WET GAP SHOWN

CARD FOOTPRINT: 85 x 55, corner R3 (business-card format).
Nominal cavity area: 4667.3 mm² including corner radii.
8 bolt positions act through metal backers and spacer frame.

GAP RULE: wet pulp gap = measured spacer - measured mesh stack.
Rings: 1.4 / 2.4 / 3.4 / 4.4 / 5.4 (one ring at a time).
Assumed compressed mesh: 0.20 per face; measure your actual mesh.
Final dry thickness shrinks and must be measured after drying.

DESIGN STATUS: dimensional manufacturing concept; loading, print quality and flatness require validation.

# P01 / PETG DRAIN PLATES / 2 IDENTICAL PARTS



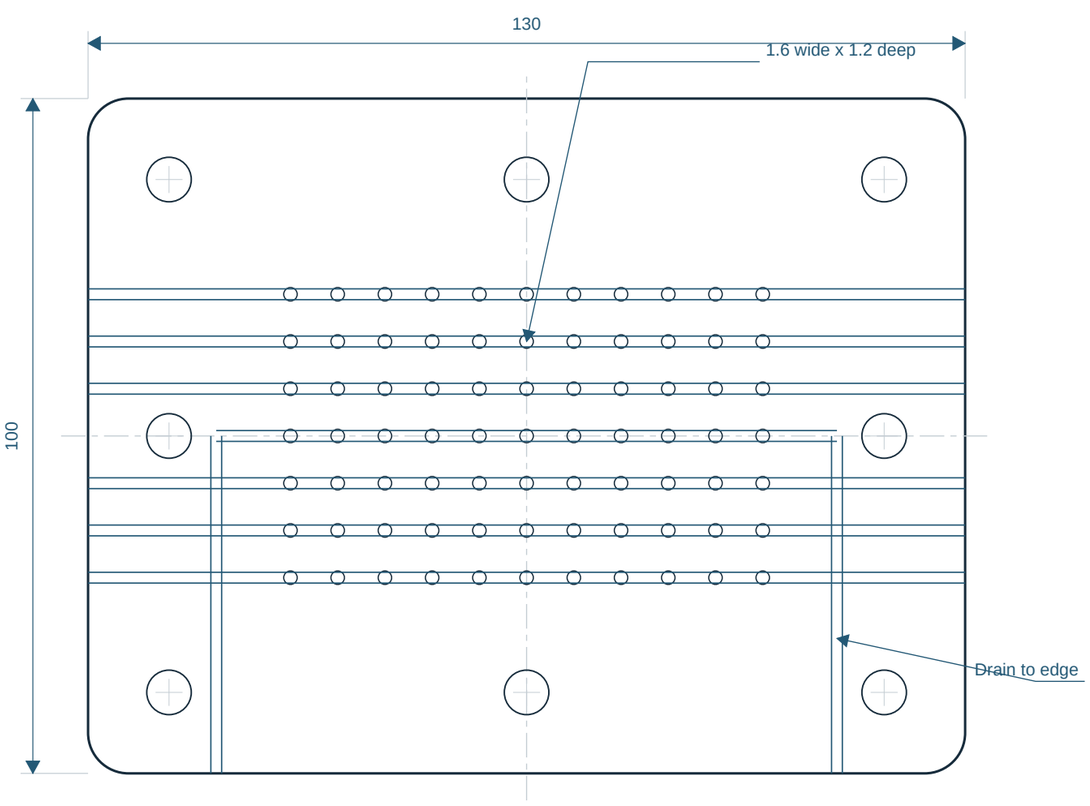

## PULP FACE / FLAT / Ø2 DRAINS

### 8 x Ø6.6 THRU / BOLT HOLE COORDINATES

Origin: centre of outer profile. X right; Y up.

| HOLE | X | Y | HOLE | X | Y |
|---|---|---|---|---|---|
| H1 | -53 | -38 | H5 | +53 | +0 |
| H2 | +0 | -38 | H6 | -53 | +38 |
| H3 | +53 | -38 | H7 | +0 | +38 |
| H4 | -53 | +0 | H8 | +53 | +38 |

Print pulp face on a flat surface; calibrate holes, remove burrs, check both faces flat. Nominal FDM XY tolerance ±0.3.

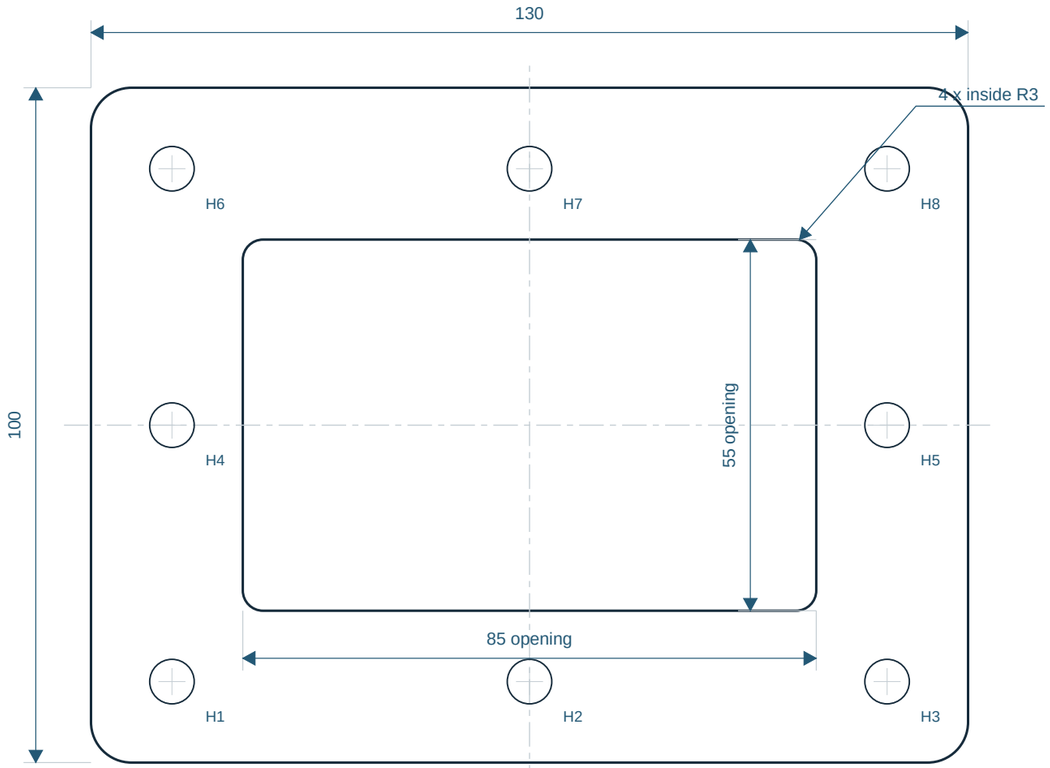



## BACK FACE / GROOVES FACE STEEL

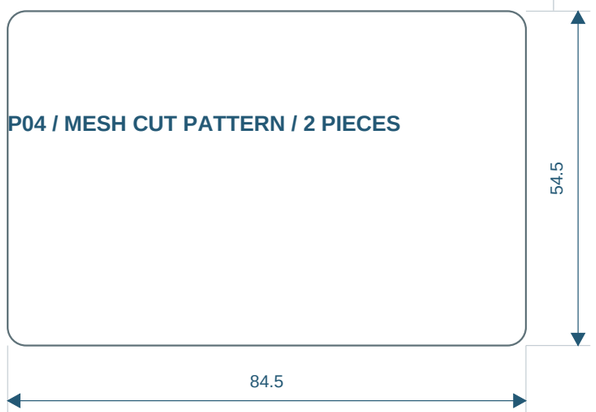
GROOVE MAP (centreline coordinates, mold XY):
6 horizontal runs: Y = ±7, ±14, ±21; X = -65 to +65.
Centre run: Y = 0; X = -46 to +46.
2 branches: X = ±46; Y = 0 to -50 (edge outlets).
Keep grooves on back face; upper plate is flipped in assembly.

DRAIN ARRAY: X = -35..+35, Y = -21..+21; pitch 7 both ways.
11 columns x 7 rows = 77 holes per plate. No blind drains.
PLATE THICKNESS 12; floor above grooves 10.8 minimum.

# P02 / INTERCHANGEABLE SPACERS + P04 / MESH




## P04 / MESH CUT PATTERN / 2 PIECES




Mesh corners R3, trim to fit freely inside opening.
Never place mesh over the ring stop faces.
100-200 µm clear aperture; buy by aperture and wire diameter / fabric thickness, then measure compressed stack.
Stainless 304/316 or polyester monofilament.

## RING THICKNESS SCHEDULE

| PART / MARK | RING (mm) | WET GAP* (mm) |
|---|---|---|
| P02-1 / G1 | 1.4 | 1.0 |
| P02-2 / G2 | 2.4 | 2.0 |
| P02-3 / G3 | 3.4 | 3.0 |
| P02-4 / G4 | 4.4 | 4.0 |
| P02-5 / G5 | 5.4 | 5.0 |

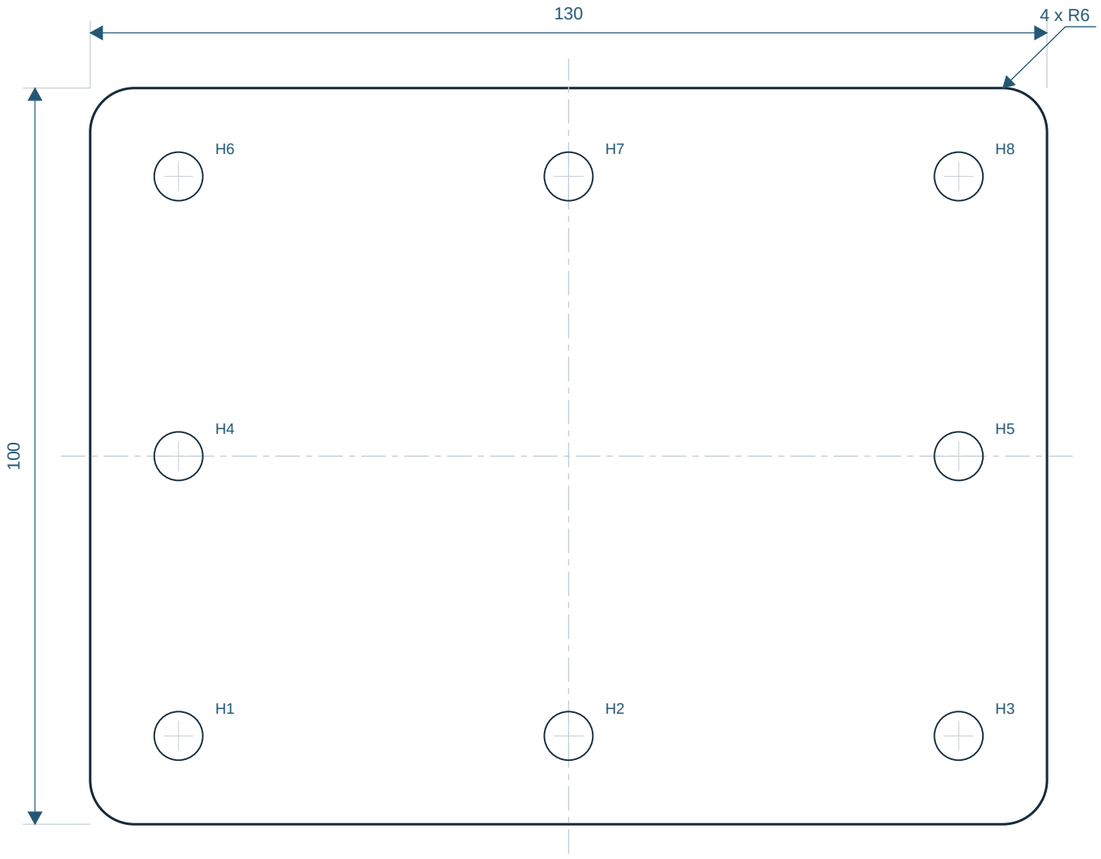
*Assumes two compressed mesh layers of 0.20 each.
Use only one continuous spacer ring per assembly.
Measure actual ring thickness at all 4 corners.
Required ring = desired wet gap + actual mesh thickness.
Do not assume wet gap equals final dry card thickness.

## SPACER SECTION (edge strip)

Horizontal flange widths: 22.5 left/right, 22.5 top/bottom.
8 x Ø6.6 THRU; same hole centres as P01 (sheet 02).
Outer profile 130 x 100 R6; cavity profile 85 x 55 R3.

FDM spacer thickness is critical: measure and lap flat as needed.

# P03 / STEEL BACKERS + FASTENER DETAILS




PLAN / 2 IDENTICAL STEEL BACKERS

EDGE VIEW / 130 x 100 x 8

## 8 x Ø6.6 THRU / BOLT HOLE COORDINATES

Origin: centre of outer profile. X right; Y up.

| HOLE | X | Y | HOLE | X | Y |
|---|---|---|---|---|---|
| H1 | -53 | -38 | H5 | +53 | +0 |
| H2 | +0 | -38 | H6 | -53 | +38 |
| H3 | +53 | -38 | H7 | +0 | +38 |
| H4 | -53 | +0 | H8 | +53 | +38 |

MATERIAL: mild steel plate, 8 thick, 2 pieces.
8 x Ø6.6 THRU; NO DRAIN HOLES OR BACKER GROOVES.
Flat faces; deburr holes and all edges.
Metal backers spread load over PETG plates.
Rounded R6 outline matches PETG parts.

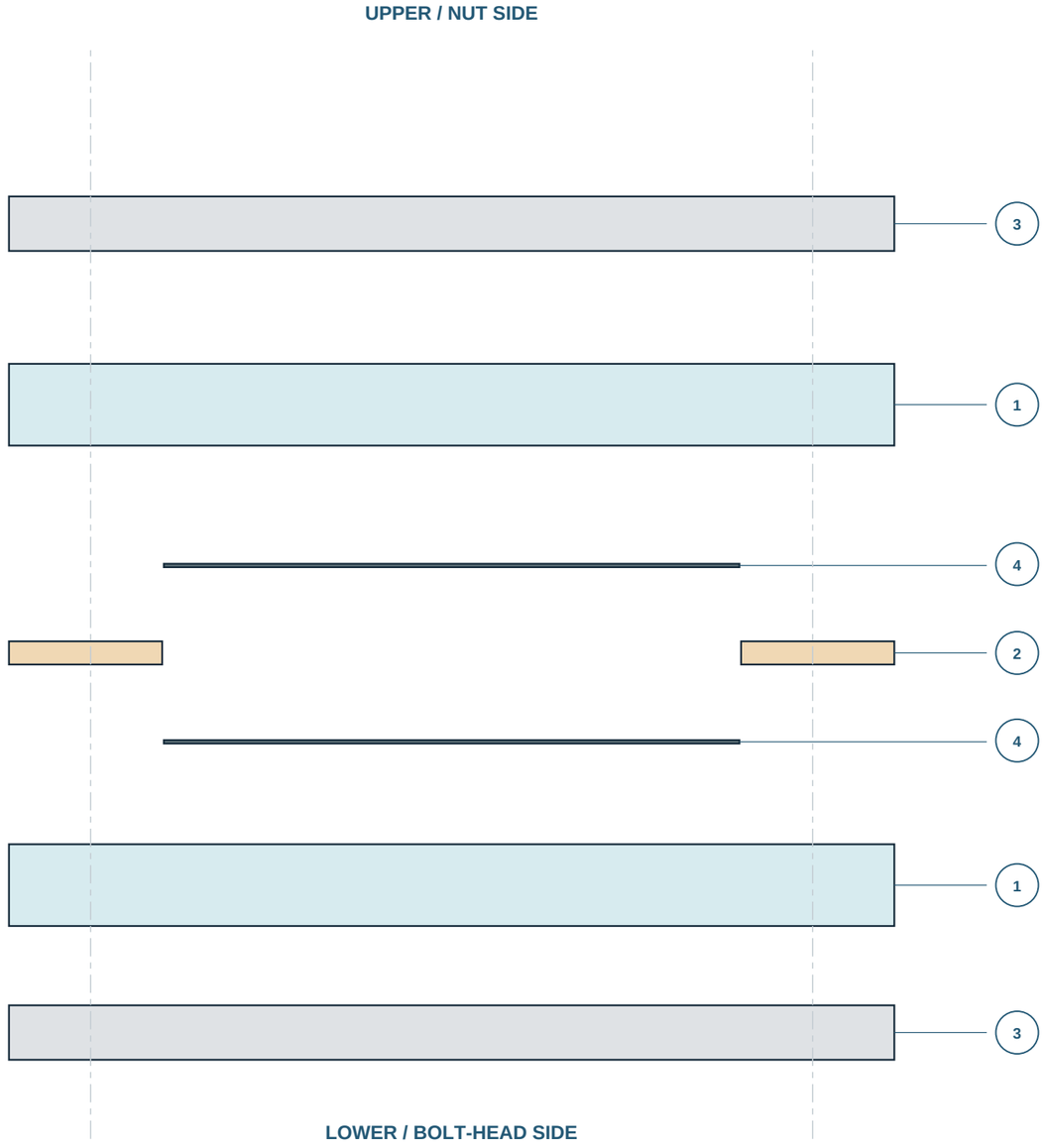
## P05-P07 / CLAMP HARDWARE

8 x M6 x 60 hex bolts, property class 8.8 suggested.
16 x M6 large-series flat washers, OD18 (ISO 7093-1).
8 x M6 nuts; plain nuts are preferred for controlled assembly.
Optional wing nuts fit M6 but do not define clamping force.
Stack: 41.4 to 45.4 (gap 1 to 5), excluding washers.
With two nominal 1.6 washers: 44.6 to 48.6.
Confirm >2 complete threads visible beyond each nut.

8 backer is recommended for this concept.
Thin aluminum/acrylic backing is not an equivalent substitution.
Bolt torque alone cannot accurately determine pulp pressure.
Use a measured press force if available; consult build guide.

FABRICATION: a workshop can cut the profile and drill holes from the 1:1 DXF or the checked A4 template (sheet 06).

# EXPLODED ASSEMBLY / BILL OF MATERIALS




Side section is exploded for assembly order; displacement is schematic.

## BILL OF MATERIALS

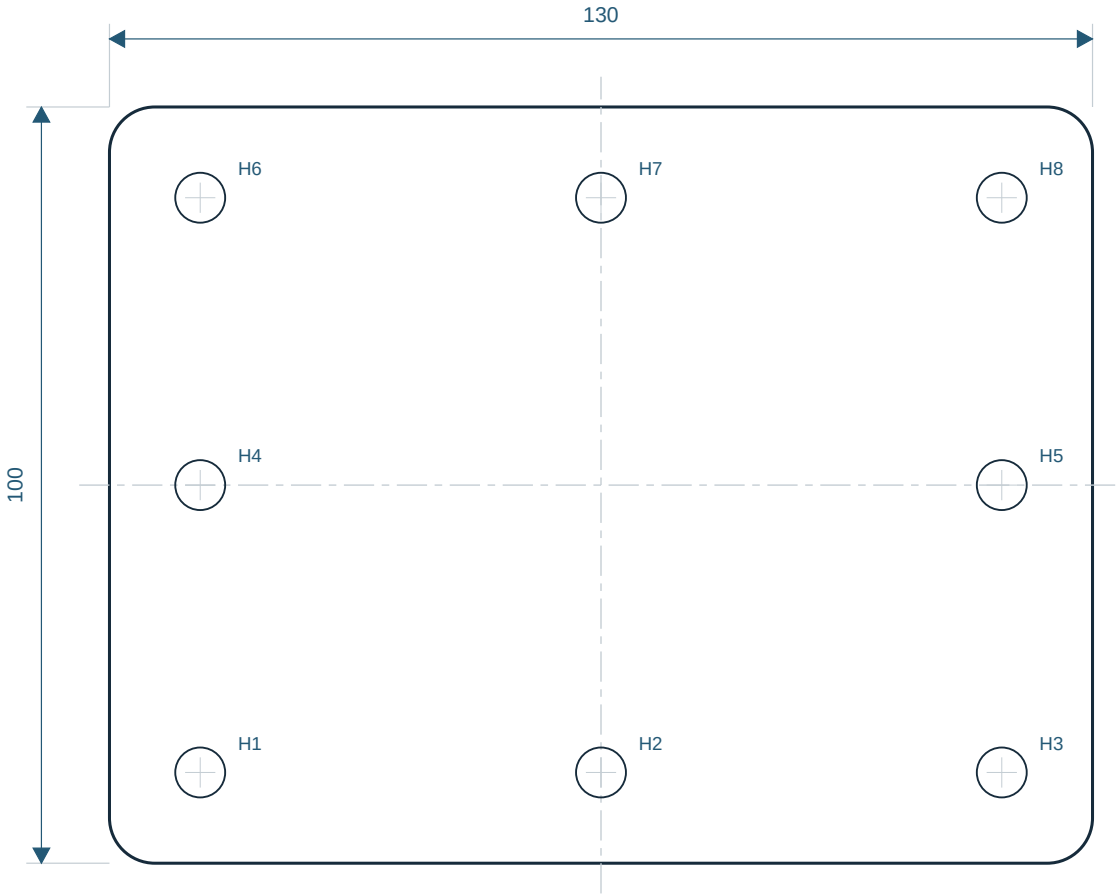
| ID | PART | QTY | DESCRIPTION |
|---|---|---|---|
| 1 | P01 | 2 | PETG drain plate 130 x 100 x 12 |
| 2 | P02 | 5 | PETG rings: 1.4, 2.4, 3.4, 4.4, 5.4 |
| 3 | P03 | 2 | Steel backer 130 x 100 x 8 |
| 4 | P04 | 2 | Filter mesh 84.5 x 54.5 R3 |
| 5 | P05 | 8 | M6 x 60 hex bolt |
| 6 | P06 | 16 | M6 washer OD18, nominal t1.6 |
| 7 | P07 | 8 | M6 nut |

ASSEMBLY ORDER:

1. Lower steel / lower PETG (grooves down) / lower mesh.
2. One selected spacer around the mesh; evenly spread wet pulp.
3. Upper mesh / upper PETG (grooves up) / upper steel.
4. Bolts through all layers; washer under each head and nut.
5. Close evenly until stops contact; dewater through mesh + drains.

MATCHED PLATES: P01 upper and lower use the same print.
SPACER SET: print all 5 to compare thickness; only 1 installed.
PULP AND MESH: confined to cavity; no fibers on stop faces.
DRYING: final dry thickness and strength require experiment.

# P03 / 1:1 STEEL DRILLING TEMPLATE




**ORIGIN (0,0) AT PROFILE CENTRE**

2 x steel backers, thickness 8.

8 x Ø6.6 THRU at shared hole table, sheet 02.

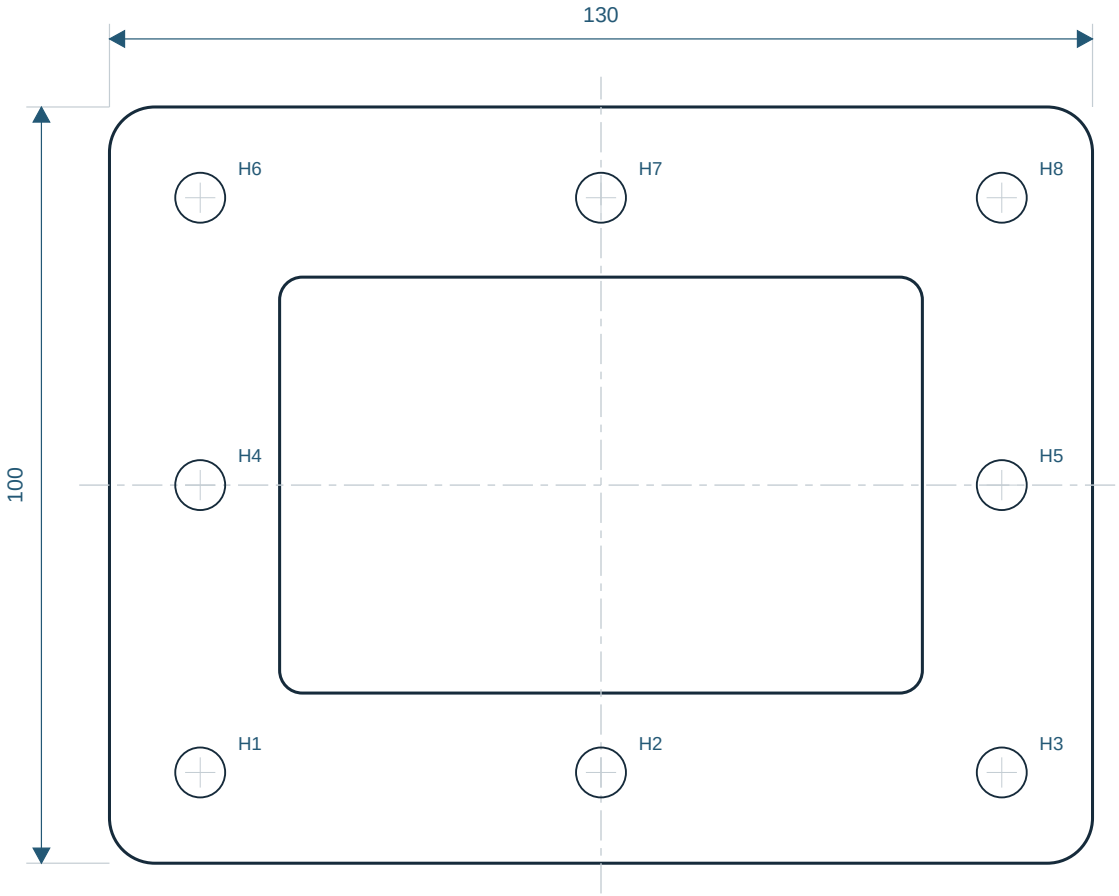
Outer contour 130 x 100; 4 corners R6.

Drill / cut only after the scale bar measures exactly 50 mm.

**CHECK: 50.00 mm**

Check bar with a ruler before transferring any dimensions.

# P02 / 1:1 SPACER PROFILE TEMPLATE




**ORIGIN (0,0) AT PROFILE CENTRE**

Outer contour 130 x 100 R6; opening 85 x 55 R3.

8 x Ø6.6 THRU at shared hole table, sheet 02.

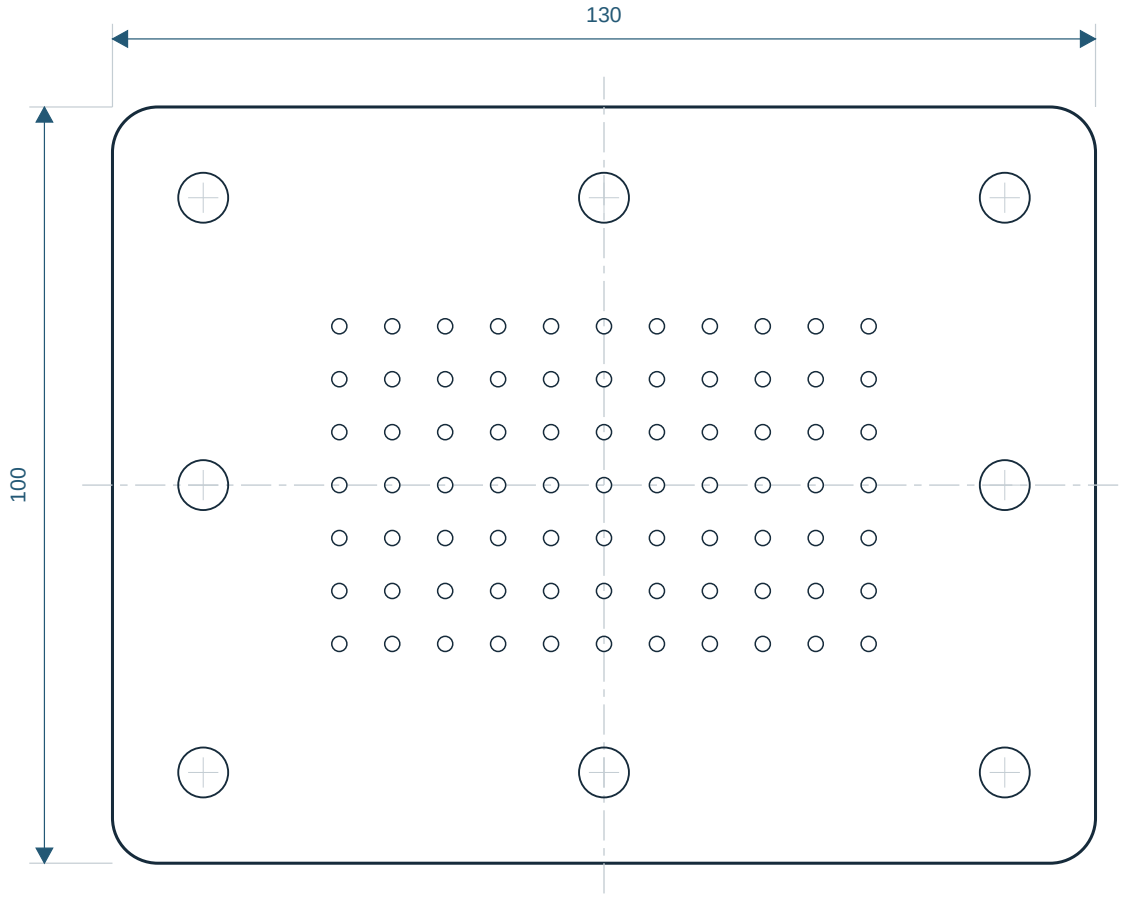
Five thicknesses: 1.4 / 2.4 / 3.4 / 4.4 / 5.4.

This is a 2D fit template; fabricate using the 3D part files.

**CHECK: 50.00 mm**

Check bar with a ruler before transferring any dimensions.

# P01 / 1:1 DRAIN FACE TEMPLATE




**ORIGIN (0,0) AT PROFILE CENTRE**

2 x PETG drain plates; thickness 12.

77 x Ø2 THRU, pitch 7; bolt holes 8 x Ø6.6 THRU.

Back channels are omitted here; see sheet 02.

This template represents the flat pulp face.

**CHECK: 50.00 mm**

Check bar with a ruler before transferring any dimensions.